FOOD COMPOSITION AND ADDITIVES

Determination of Total Dietary Fiber (CODEX Definition) by Enzymatic-Gravimetric Method and Liquid Chromatography: Collaborative Study

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A method for the determination of total dietary fiber (TDF), as defined by the CODEX Alimentarius, was validated in foods. Based upon the principles of AOAC *Official Methods*SM 985.29, 991.43, 2001.03, and 2002.02, the method quantitates high- and low-molecular-weight dietary fiber (HMWDF and LMWDF, respectively). In 2007, McCleary described a method of extended enzymatic digestion at 37°C to simulate human intestinal digestion followed by gravimetric isolation and quantitation of HMWDF and the use of LC to quantitate low-molecular-weight soluble dietary fiber (LMWSDF). The method thus quantitates the complete range of dietary fiber components from resistant starch (by utilizing the digestion conditions of AOAC Method 2002.02) to digestion resistant oligosaccharides (by incorporating the deionization and LC procedures of AOAC Method 2001.03). The method was evaluated through an AOAC collaborative study. Eighteen laboratories participated with 16 laboratories returning valid assay data for 16 test portions (eight blind duplicates) consisting of samples with a range of traditional dietary fiber, resistant starch, and nondigestible oligosaccharides. The dietary fiber content of the eight test pairs ranged from 11.57 to 47.83%. Digestion of samples under the conditions of AOAC Method 2002.02 followed by the isolation and gravimetric procedures of AOAC Methods 985.29 and 991.43 results in quantitation of HMWDF. The filtrate from the quantitation of HMWDF is concentrated, deionized, concentrated again, and analyzed by LC to determine the LMWSDF, i.e., all nondigestible oligosaccharides of degree of polymerization ≥3. TDF is calculated as the sum of HMWDF and LMWSDF. Repeatability standard deviations (s_r) ranged from 0.41 to 1.43, and reproducibility standard deviations (s_R) ranged from 1.18 to 5.44. These results are comparable to other official dietary fiber methods, and the method is recommended for adoption as Official First Action.

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Dramatic increases in the use of fiber analysis in fiber research, and the marketing, research, and development of food products have accompanied increases in public awareness of the health benefits of high

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fiber foods over the past several decades. As researchers have discovered and elucidated additional dietary fiber sources, there has been a need not only to update the definition of dietary fiber, but also to update the methodologies that support the definition. AOAC INTERNATIONAL has been a leader in providing *Official Methods of Analysis*SM consistent with the state of dietary fiber science (1). In the 1970s, Trowell and fellow dietary fiber researchers (2–5) published a definition later adopted as consensus by AOAC INTERNATIONAL following an international survey by Prosky et al. (6) in the late 1970s.

The Trowell et al. (5) definition.—Dietary fiber consists of the plant polysaccharides and lignin which are resistant to hydrolysis by digestive enzymes of man. This definition defines a macro constituent of foods which includes cellulose, hemicellulose, lignin, gums, modified celluloses, mucilages, oligosaccharides, and pectins and associated minor substances such as waxes, cutin, and suberin.

As the science of dietary fiber has advanced, AOAC *Official Methods* **985.29** (7) and its extensions **991.42** and **993.19** (8) and methods which give equivalent results, **991.43**, **992.16**, **993.21**, and **994.13** (9), which adequately quantitated the known dietary fiber food fractions at the time of the Trowell et al. definition, have become insufficient to quantitate other dietary fiber components which were unknown at the time of adoption of those *Official Methods*. Advances in understanding the complexity of dietary fiber, and that food components such as resistant starch (RS), fructans, polydextrose, and resistant maltodextrins are part of dietary fiber in the diet led to the development and adoption of *Official Methods* **997.08** and **999.03**, **2000.11**, **2001.02**, **2001.03**, and **2002.02** (for the quantitation of fructans, polydextrose, *trans*-galacto-oligosaccharides, resistant maltodextrins, and RS, respectively). Although these methods perform well for their individual components, obtaining a total dietary fiber (TDF) value involves performing multiple analyses and mathematical summations, and in some cases carrying out special procedures to ensure against double counting of a particular dietary fiber fraction (10).

Recently, international authorities on dietary fiber definition, working through the CODEX Committee on Nutrition and Foods for Special Dietary Uses (CCNFSDU), have updated the terminology of the dietary fiber definition (11).

CODEX defines dietary fiber as carbohydrate polymers (**a**) with 10 or more monomeric units (**b**), which are not hydrolyzed by the endogenous enzymes in the small intestine of humans and belong to the following categories: Edible carbohydrate polymers naturally occurring in the food as consumed; carbohydrate polymers, which have been obtained from food raw material by physical, enzymatic, or chemical means and which have been shown to have a physiological effect of benefit to health as demonstrated by generally accepted scientific evidence to competent authorities; and synthetic carbohydrate polymers which have been shown to have a physiological effect of benefit to health as demonstrated by generally accepted scientific evidence to competent authorities.

(**a**) Dietary fiber derived from a plant origin may include fractions of lignin and/or other compounds when associated with polysaccharides in the plant cell walls and if these compounds are quantified by the AOAC gravimetric analytical method for dietary fiber analysis; e.g., fractions of lignin and the other compounds (proteic fractions, phenolic compounds, waxes, saponins, phytates, cutin, phytosterols, etc.) intimately associated with plant polysaccharides in AOAC Method **991.43**. These substances are included in the definition of fiber as they are actually associated with the poly- or oligosaccharidic fraction of fiber. However, when extracted or even reintroduced into a food containing nondigestible polysaccharides, they cannot be defined as dietary fiber. When combined with polysaccharides, these associated substances may provide additional beneficial effects (pending adoption of Section on Methods of Analysis and Sampling).

(**b**) Decision on whether to include carbohydrates of 3 to 9 monomeric units should be left up to national authorities.

With the possible exception of the synthetic carbohydrate polymers, the food components included in the CODEX definition match those of the more broadly stated Trowell et al. (5) definition. Therefore, the currently adopted Official Methods can be readily applied, along with the same issues of extra expense for multiple assays, the need for mathematical summations, and special procedures needed to avoid double counting. Obviously, a single method that quantitates all of the components included in the Trowell et al. and the CODEX definitions is desired. Because the CODEX definition includes resistant oligosaccharides of the size of 3 monomeric units and larger (as had previous proposals agreed upon within the CODEX committee), a relevant method must meet that criteria. If a demarcation between degree of polymerization (DP) 9 and DP 10 is necessary, a methodology will have to be worked out if that national authority option is pursued. This may prove to be a difficult task. According to Ku et al. (12), some carbohydrate polymers of DP <10 may remain in the alcoholic precipitate, whereas for other carbohydrates, polymers of DP >10 are not precipitated. Therefore, there is currently no clear-cut approach to quantitation at a demarcation point between DP 9 and DP 10.

In 2007, McCleary (13) described a method that incorporates key features of *Official Methods* **985.29**, **991.43**, **2001.03**, and **2002.02** (14, 15) to provide an assay that quantitates the food fraction as delineated by CCNFSDU. This methodology addresses the issues expressed above, and provides a method that is based on methodologies accepted by industries and authorities worldwide. One remaining concern with the inclusive method is that the entire process of analysis takes 2 days. The all-inclusive nature of the most recent methods has increased the time relative to Methods **985.29** and **991.43**. However, this is necessary to ensure quantitation of all fiber fractions, and it consumes less time than carrying

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Table 1. Results of precollaborative ruggedness testing (% TDF reported by laboratories)

Lab No./sample	Whole grain bread	Defatted peanuts	Haricot beans	Resistant starch	Defatted soybean	Low-carbohydrate spaghetti
1	6.23[a]	14.13	51.05	43.19	23.51	4.64
2	10.58	12.51	50.69	41.24	21.82	6.75
3	11.45	11.45	50.25	42.70	23.70	7.70[a]
4	9.75	15.21	45.46[a]	41.50	21.03	4.70
5	11.13	16.43[a]	51.27	41.35	24.06	5.92
6	10.05	14.16	49.33	41.69	21.43	5.18
7	10.25	18.76[a]	51.23	46.20[a]	21.16	9.70[a]
8	9.86	14.67	47.85	41.56	20.59	4.80
9	9.97	12.37	47.63	41.53	18.96[a]	4.38
10	13.94[a]	13.22	52.42	44.64	22.19	5.25
11	10.53	13.73	51.01	44.01	23.13	5.43
12	10.33	14.25	50.34	38.61[a]	22.42	5.61
13	8.43[a]	23.33[a]	48.19	39.99	23.29	4.72
14	9.80	14.00	52.50	43.40	22.90	5.10
15	8.85[a]	12.36	49.27	43.21	15.41[a]	3.04[a]
Average	10.33	13.51	50.22	42.31	22.40	5.21
S_R	0.53	1.12	1.57	1.32	1.13	0.66
RSD_R, %	5.11	8.30	3.12	3.12	5.04	12.66

[a] Statistical outlier.

out several assays separately so that the results can be mathematically combined.

Collaborative Study

Precollaborative Ruggedness Testing

A precollaborative ruggedness study was conducted with a number of laboratories to ensure adequate method performance. A call was placed for volunteer laboratories to participate in evaluating the methodology. Volunteer laboratories were sent six samples along with copies of the method, ion exchange resins, and a supply of enzymes. Each laboratory was requested to run each sample in singlet. Laboratories were requested to conduct the analysis, ask questions regarding procedures and write-up, and provide feedback to the Study Directors on any aspects of the method about which the collaborator might have a concern. The results of the analysis on the ruggedness testing samples are shown in Table 1. Relevant comments received from the participating laboratories were incorporated as changes to improve the method as appropriate. No procedural changes were found to be necessary, and only minor edits of the text for clarity were made.

The results of the precollaborative ruggedness study were typical for dietary fiber methods. Repeatability, reproducibility, and the HorRat values were within the range of performance characteristics typically found for dietary fiber methods, wherein a significant number of manual steps are necessary to carry out the assay. The between-laboratory variability s_R ranged from 0.53 to 1.57 for TDF, and the between-laboratory relative variability RSD_R ranged from 3.12 to 12.66%. This is due to the fact that all dietary fiber methods to date comprise a significant number of technique-dependent manual operations, each of which contributes to the overall variability of the final results. By way of comparison, the statistical characteristics of the various AOAC *Official Methods* of analysis are compiled in Table 2. The statistical characteristics of this method, which combines steps from AOAC *Official Methods* **985.29**, **991.43**, **2001.03**, and **2002.02**, lie within the ranges of the statistical characteristics of the current *Official Methods*.

Therefore, the Study Directors determined that the method was ready for full collaborative study.

Collaborative Study Protocol

Eight food samples were selected for the collaborative study. Because the method under consideration incorporates RS and nondigestible oligosaccharides into a more traditional dietary fiber methodology, the samples for this collaborative study were chosen to be challenging, i.e., with emphasis on quantitating products high in RS (legumes, RS ingredient, and whole grain products) and products with typical levels of nondigestible oligosaccharides (all samples). Methods designed to quantitate more traditional dietary fiber have been thoroughly studied and validated since 1980 (e.g., AOAC **985.29**, **991.43**, etc). Inclusion of components such as RS and

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Table 2. Comparable AOAC Method data

Method No.	Title of method	S_r	RSD_r	S_R	RSD_R	HorRat
985.29	Total Dietary Fiber in Foods	0.15–0.99	0.56–66.25	0.27–1.36	1.58–66.25	0.76–17.46
991.42	Insoluble Dietary Fiber in Food and Food Products	0.41–2.82	0.86–10.38	0.62–9.49	3.68–19.44	1.73–8.68
991.43[a]	Insoluble Dietary Fiber in Food and Food Products	0.36–1.06	1.50–6.62	0.85–2.06	1.58–12.17	0.74–4.66
992.16	Total Dietary Fiber	0.18–1.01	1.48–14.73	0.22–2.06	4.13–17.94	1.84–4.62
993.19	Soluble Dietary Fiber in Food and Food Products	0.49–1.15	1.74–5.93	0.79–2.05	2.41–7.01	1.13–2.83
994.13	Total Dietary Fiber (Determined as Neutral Sugar Residues, Uronic Acid Residues, and Klason Lignin)	0.32–2.88	1.80–6.96	0.52–4.90	4.80–11.30	2.32–4.20
2001.03	Dietary Fiber Containing Supplemented Resistant Maltodextrin (RMD)	0.02–1.63	1.33–6.10	0.04–2.37	1.79–9.39	0.77–3.32
2002.02	Resistant Starch in Starch and Plant Materials	0.08–2.66	1.97–4.12	0.21–3.87	4.58–10.9	1.44–3.74

[a] Samples that were not dried and/or desugared only.

nondigestible oligosaccharides in the CODEX Alimentarius definition indicates that updated testing procedures must include the capability of accurately quantitating these components.

Moist samples were freeze-dried before grinding. All samples were ground to method-specific size and homogenized by thorough mixing before being subdivided into polyethylene bottles and sealed. Samples, copies of the method, report sheets, and sample storage instructions, along with an adequate supply of enzymes and deionizing resins were shipped to collaborating laboratories by express overnight shipment.

A total of 18 laboratories reported data for the collaborative study samples. One laboratory reported little or no low-molecular-weight soluble dietary fiber (LMWSDF) for their assays. Subsequent investigation revealed an ion exchange resin had undergone a color change; however, the reason for loss of LMWSDF was not fully determined. Another laboratory used high-pressure anion exchange chromatography with electrochemical detection instead of the prescribed LC method. Data from these two laboratories are not included in data tables or statistical analysis.

Statistical Treatment

Collaborating laboratory data were evaluated statistically according to AOAC protocols using AOAC-supplied software. Of the 128 valid pairs of assay results reported, Laboratories 1–5, 8, 9, 11, 13, and 16 had no statistical outliers; Laboratories 6, 12, 14, and 15 had one statistical outlier; Laboratory 7 had three statistical outliers; and Laboratory 10 had four statistical outliers, for a total of 11 statistical outliers overall. The data from Laboratories 17 and 18 were not included in the statistical evaluation due to operational issues as reported by those laboratories. The paired data from the blind duplicate results reported by the collaborating laboratories are shown in Table 3. Outliers, and their reason for removal, are indicated in Table 3.

AOAC Official Method 2009.01
Total Dietary Fiber in Foods

Enzymatic-Gravimetric-Liquid Chromatographic Method
First Action 2009

[Applicable to plant material, foods, and food ingredients consistent with CODEX Definition 2008 (ALINORM 09/32/26), including naturally occurring, isolated, modified, and synthetic polymers meeting that definition.]

See Table **2009.01** for the results of the interlaboratory study supporting acceptance of the method.

A. Principle

A method is described for the measurement of total dietary fiber (TDF), including resistant starch (RS) and low-molecular-weight nondigestible oligosaccharides (LMWNDO) of degree of polymerization (DP) ≥ 3 (Figure **2009.01A**). The method combines the key attributes of AOAC *Official Methods* **985.29**, **991.43**, **2001.03**, and **2002.02** (Figure **2009.01B**). Duplicate test portions are incubated with pancreatic α-amylase and amyloglucosidase (AMG) for 16 h at 37°C in sealed 250 mL bottles in a shaking water bath while mixing with sufficient vigor to maintain continuous suspension. During this step, nonresistant starch is solubilized and hydrolyzed to glucose and maltose by the combined action of the two enzymes. The reaction is terminated by pH adjustment and temporary heating. Protein in the sample is digested with protease. For the measurement of high-molecular-weight dietary fiber (HMWDF), ethanol or industrial methylated spirits (IMS) are added, and the insoluble and precipitable soluble dietary fiber is captured, washed with ethanol and acetone, dried, and weighed. One of the duplicate residues is analyzed for protein, and the other for ash. Nonprecipitable dietary fiber in the filtrate is recovered by concentrating, then desalting through ion exchange resins, and concentrating and quantitating by LC.

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Table 3. Interlaboratory study data for total dietary fiber (% TDF reported by laboratories)

Sample/ Lab No.	Bran cereal		Broccoli, freeze-dried		Carrots, freeze-dried		Haricot beans		Resistant starch		Red kidney beans		Whole grain bread		Whole grain pasta	
1	31.60	31.54	30.38	30.25	25.94	24.94	50.34	50.40	45.86	46.10	24.87	24.05	12.21	11.85	13.67	12.99
2	30.64	31.25	30.44	30.14	24.89	25.64	48.26	46.80	47.16	47.76	24.10	25.06	11.82	13.30	12.45	14.19
3	30.78	34.47	33.74	32.60	26.66	25.87	49.73	50.02	45.09	42.02	27.30	27.25	11.87	15.04	13.52	12.30
4	29.79	29.23	28.31	28.11	23.08	23.09	33.07	34.89	48.63	49.96	24.17	24.82	11.81	11.21	12.35	11.92
5	29.10	29.80	27.60	28.20	23.60	23.80	48.90	49.60	42.90	42.10	22.20	21.70	10.00	10.10	11.10	10.80
6	32.46	34.38	32.12	33.57	25.81	26.83	53.29	53.79	54.28	52.49	25.57	26.28	14.41	14.04	15.60[a]	15.52[a]
7	36.61	34.62	29.66	30.55	24.28	23.74	54.90	52.49	49.34	50.57	32.16[b]	24.10[b]	33.42[b]	16.95[b]	36.43[b]	13.39[b]
8	30.83	30.04	29.34	29.08	26.10	26.20	42.39	40.11	44.76	40.80	23.51	23.76	12.21	10.86	11.34	12.39
9	28.23	29.11	29.52	30.09	24.11	24.16	47.56	48.72	45.11	44.71	23.44	23.29	12.73	11.34	12.13	11.36
10	33.61[b]	24.51[b]	27.82	26.22	20.27[c]	18.97[c]	41.06	42.76	34.06	34.46	22.71[b]	17.21[b]	12.14	7.14	5.01[b]	11.81[b]
11	31.97	32.29	30.93	30.91	25.98	26.15	50.14	49.85	45.54	45.84	25.78	25.52	14.34	13.36	13.86	13.35
12	29.65	28.65	30.11	32.52	[d]	20.22[c]	48.20	46.97	36.31	40.50	21.31	20.95	5.69	9.77	4.19	3.82
13	32.33	31.69	32.75	30.35	25.68	25.00	56.40	51.93	44.80	44.72	29.24	28.31	11.21	9.94	11.65	11.49
14	29.22	29.35	28.95	28.92	24.03	23.41	38.04[b]	48.80[b]	35.87	35.81	21.73	22.17	10.16	10.31	10.17	11.14
15	32.69	31.61	31.98	31.83	27.74[b]	25.15[b]	44.48	47.84	43.24	44.44	24.73	23.91	13.76	13.47	14.42	14.07
16	31.60	31.78	33.07	33.30	25.80	25.82	50.37	49.50	42.74	42.79	23.70	24.00	10.28	10.61	12.60	12.50

[a] Single Grubbs test outlier on total dietary fiber.
[b] Cochran test outlier on total dietary fiber.
[c] Double Grubbs test outlier on total dietary fiber.
[d] Laboratory reported no result for this sample.

B. Apparatus

(**a**) *Grinding mill*.—Centrifugal, with 12-tooth rotor and 0.5 mm sieve, or similar device. Alternatively, a cyclone mill can be used for small test laboratory samples provided the mill has sufficient air flow or other cooling to avoid overheating samples.

(**b**) *Digestion bottles*.—250 mL Duran® glass bottles with plastic caps or 250 mL polypropylene bottles with polypropylene caps.

(**c**) *Fritted crucible.*—Büchner, fritted disk, Pyrex® 60 mL, pore size, coarse, American Society for Testing and Materials (ASTM) 40–60 µm, Corning No. 36060, or equivalent. Prepare as follows:

(*1*) Ash overnight at 525°C in muffle furnace; cool furnace to 130°C before removing crucibles to minimize breakage. (*2*) Remove any residual Celite and ash material by using a vacuum. (*3*) Soak in 2% cleaning solution, **C**(**i**), at room temperature for 1 h. (*4*) Rinse crucibles with water and deionized water. (*5*) For final rinse, use 15 mL acetone and air-dry. (*6*) Add approximately 1.0 g Celite to dried crucibles and dry at 130°C to constant weight. (*7*) Cool crucible in desiccators for approximately 1 h and record mass of crucible containing Celite.

(**d**) *Filtering flask.*—Heavy-walled, 1 L with side arm.

(**e**) *Rubber ring adaptors*.—Used to join crucibles with filtering flasks.

(**f**) *Vacuum source*.—Vacuum pump or aspirator with regulator capable of regulating vacuum.

(**g**) *Water bath(s)*.—Rotary motion or horizontal shaking, large-capacity (20–24 L) with covers; capable of maintaining temperatures of 37 ± 1°C and 60 ± 1°C; equipped with automatic timers for on–off operation or equivalent (e.g., Grant® OLS 200 shaking incubation bath). Ensure that shaking action/sample agitation in water bath used is sufficient to maintain sample solids in suspension, and no residue buildup or rings of sample material form in the digestion bottle during the enzymatic digestions; for example, a back-and-forth shaker with the bottles placed at a 45° angle will provide adequate agitation, whereas placing the bottle vertically or horizontally will not.

(**h**) *Balance*.—0.1 mg readability, accuracy, and precision.

(**i**) *Ovens*.—Two, mechanical convection, set at 103 ± 2°C and 130 ± 3°C.

(**j**) *Timer.*

(**k**) *Desiccator*.—Airtight, with SiO_2 or equivalent desiccant. Desiccant dried biweekly overnight in 130°C oven.

(**l**) *pH meter.*

(**m**) *Pipettors and tips.*—50–200 µL and 5 mL capacity.

(**n**) *Dispensers.*—(*1*) 15 ± 0.5 mL for 78% EtOH (or IMS), 95% ethanol (or IMS), and acetone; (*2*) 40 ± 0.5 mL for buffer.

(**o**) *Cylinder*.—Graduated, 500 mL.

(**p**) *Magnetic stirrers and stirring bars.*

(**q**) *Rubber spatulas.*

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Table 2009.01. Interlaboratory study results for total dietary fiber in foods

Sample/ parameter	Bran cereal	Broccoli, freeze-dried	Carrots, freeze-dried	Haricot beans	Resistant starch	Red kidney beans	Whole grain bread	Whole grain pasta
No. labs	15	16	13	15	16	14	15	13
Mean, %	31.24	30.42	25.02	47.83	44.09	24.38	11.57	12.65
S_r	0.94	0.78	0.41	1.36	1.28	0.42	1.43	0.57
S_R	2.00	1.95	1.18	5.44	5.05	2.10	2.08	1.43
RSD_r	3.01	2.55	1.65	2.83	2.89	1.71	12.34	4.47
RSD_R	6.42	6.42	4.70	11.38	11.45	8.6	17.97	11.31
HorRat	2.69	2.68	1.91	5.09	5.06	3.48	6.49	4.14

(**r**) *Muffle furnace*.—525 ± 5°C.

(**s**) *Glass or polypropylene columns, 20 cm × 2.5 cm id*.—To hold ion exchange resins with fittings and plastic tubing for filling and draining.

(**t**) *LC system*.—With oven to maintain a column temperature of 90°C and a 50 µL injection loop. Column operating conditions: Temperature, 90°C; mobile phase, distilled water plus $Na_2CaEDTA$ (50 mg/L); flow rate, 0.5 mL/min for Sugar-Pak column. System must separate maltose from higher malto-oligosaccharides. Run time of 30 min to ensure column is cleaned out. Use only distilled water if size exclusion columns are used. System must separate maltose from maltotriose [Figures **2009.01C**(**a**) and (**b**) and **2009.01D**(**a**) and (**b**)]. Run time of 60 min to ensure columns are cleaned out.

(**u**) *Guard column (or precolumn)*.—Waters Guard Pak LC precolumn inserts (Waters, Milford, MA) Part No. WAT015209, or equivalent.

(**v**) *LC columns.—Option 1: Waters Sugar-Pak*.—6.5 × 300 mm (Part No. WAT085188), or equivalent. *Option 2: Size exclusion LC columns*.—Two TSK-Gel 30 cm × 7.8 mm connected in series, G2500PWXL (Sigma Aldrich, St. Louis, MO) Part No. 808020).

(**w**) *Detector*.—Refractive index (RI); maintained at 50°C.

(**x**) *Data integrator or computer*.—For peak area measurement.

(**y**) *Filters for disposable syringe*.—Polyvinylidene fluoride 0.45 µm, 13 or 25 mm.

(**z**) *Filters for water*.—Polyvinylidene fluoride, 0.45 µm, 47 mm.

(**aa**) *Filter apparatus*.—To hold 47 mm, 0.45 µm filter, (**z**); to filter larger volumes of water.

(**bb**) *Syringes*.—10 mL, disposable, plastic.

(**cc**) *Syringes*.—Hamilton® 710SNR; 100 µL.

(**dd**) *Rotary evaporator*.—Heidolph Laborota® 4000, or equivalent.

(**ee**) *Thermometer*.—Capable of measuring to 110°C.

C. Reagents

(**a**) *Ethanol 95% (v/v) or IMS*.—IMS consist of ethanol 84.83 (w%), 85.95 (v%); water 5.66 (w%), 4.52 (v%); 2-propanol 4.91 (w%), 5.00 (v%); methanol 4.60 (w%), 4.52 (v%). Prepare by mixing 5 volumes 2-propanol with 95 volumes denatured ethanol formula SDA-3A (100 volumes 95% ethanol combined with 5 volumes methanol).

(**b**) *Ethanol (or IMS), 78%*.—Place 207 mL water into 1 L volumetric flask. Dilute to volume with 95% ethanol or IMS. Mix.

(**c**) *Acetone*.—Reagent grade.

(**d**) *Stock AMG solution, 3300 units/mL in 50% (v/v) glycerol*.—Solution is viscous; for dispensing use positive displacement dispenser. AMG solution is stable for up to 5 years when stored at 4°C. (*Note*: One unit of enzyme activity is amount of enzyme required to release 1 µmole glucose from soluble starch/min at 40°C and pH 4.5.) AMG solution should be free of detectable levels of free glucose.

(**e**) *Pancreatic α-amylase (50 units/mL)/AMG (3.4 units/mL)*.—Immediately before use, dissolve 0.10 g purified porcine pancreatic α-amylase (150 000 units/g; Ceralpha method; AOAC Method **2002.01**) in 290 mL sodium maleate buffer (50 mM, pH 6.0, plus 2 mM $CaCl_2$ and 0.02% sodium azide) and stir for 5 min. Add 0.3 mL AMG [**C**(**d**)].

(**f**) *Protease (50 mg/mL; approximately 350 tyrosine units/mL) in 50% (v/v) glycerol*.—Solution is viscous; for dispensing use positive displacement dispenser. Use as supplied.

(**g**) *Deionized water*.

(**h**) *Celite*.—Acid-washed, pre-ashed (Megazyme G-CEL100 or G-CEL500).

(**i**) *Cleaning solution*.—Micro-90® (International Products Corp., Trenton, NJ). Make a 2% solution with deionized water.

(**j**) *Sodium maleate buffer*.—50 mM, pH 6.0 plus 2 mM $CaCl_2$ and 0.02% sodium azide. Dissolve 11.6 g maleic acid in 1600 mL distilled water and adjust pH to 6.0 with 4 M (160 g/L) NaOH solution. Add 0.6 g calcium chloride ($CaCl_2 \cdot 2H_2O$) and 0.4 g sodium azide. [*Note*: Do not add the sodium azide until the pH has been adjusted. Acidification of sodium azide releases a poisonous gas. Handle sodium azide and maleic acid with caution only after reviewing Material Safety Data Sheet (MSDS), using appropriate personal

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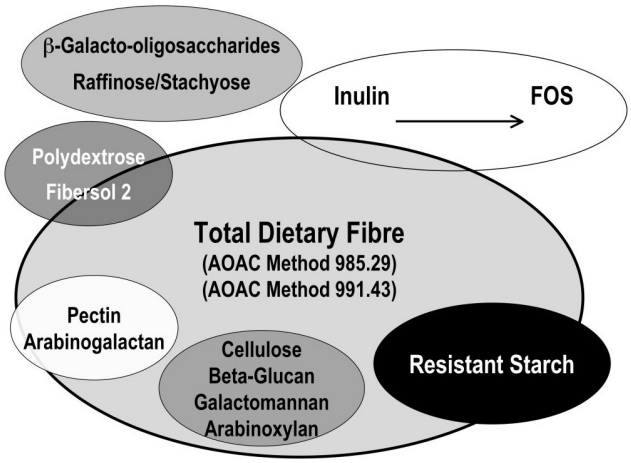

Figure 2009.01A. Schematic representation of dietary fiber components measured by AOAC Official Methods 985.29 and 991.43. Also depicted are the problems of partial measurement of RS, polydextrose, and resistant maltodextrins by current AOAC dietary fiber methods. Most of the LMWSDF (galacto-oligosaccharides, fructo-oligosaccharides, etc.) are not measured. The current integrated total dietary fiber procedure measures all components shown, with no possibility of double counting.

protective gear and laboratory hood and adjust the volume to 2 L. Stable for >1 year at 4°C.]

(**k**) *Trizma base (Sigma Cat. No. T-1503), 0.75 M.*—Add 90.8 g Trizma base to approximately 800 mL distilled water, and dissolve. Adjust volume to 1 L. Stable for >1 year at room temperature.

(**l**) *Acetic acid solution, 2 M.*—Add 115 mL glacial acetic acid (Fluka 45731) to a 1 L volumetric flask. Dilute to 1 L with distilled water.

(**m**) *pH standards.*—Buffer solutions at pH 4.0, 7.0, and 10.0.

(**n**) *LC retention time standard.*—Standard source having the distribution of oligosaccharides (DP ≥3) corn syrup solids (DE 25; Matsutani Chemical Industry Co., Ltd, Itami City, Hyogo, Japan; www.matsutani.com), analyzed by LC. Dissolve 2.0 g portion of oligosaccharide mixture and 0.50 g maltose in deionized water and transfer to 100 mL volumetric flask. Pipet 10 mL internal standard, **C**(**p**). Dilute to volume with 0.02% sodium azide solution, **C**(**t**). Transfer solutions to 100 mL polypropylene bottle. Stable at room temperature for 1 year. When size exclusion chromatography is used, sorbitol is replaced by the same mass of diethylene glycol. Alternatively, diethylene glycol can be used with both systems (not used in the current study).

(**o**) *Mixed-bed ion exchange resins for each test portion.*—(*1*) *m-1.*—25 g Amberlite FPA53 (OH^-) resin (Rohm and Haas S.A.S., Paris, France), ion exchange capacity (R-OH exchange capacity data supplied by manufacturer) 1.6 meq/mL (min) or equivalent, and (*2*) *m-2.*—25 g Ambersep 200 (H^+) resin or equivalent (Rohm and Haas S.A.S.) are mixed together just prior to use and packed in column, **B**(**s**), for analysis of each test portion. The converted resin should satisfy the following specifications: Ion exchange capacity (R-H exchange capacity data supplied by manufacturer) 1.6 meq/mL (min), pH 4–7. Before mixing and packing the two resins into a column, wash each resin with H_2O to obtain a pH value of 7–8 for *m-1* and 4–7 for *m-2*. *Note*: If using Amberlite 200C "Na-type" resin, convert into the H^+ form by mixing 500 mL resin with 2 L 1 M HCl in a 5 L beaker. Swirl the suspension occasionally over a 1 h period, allow the resin to settle, and then decant the supernatant solution. Add 4 L distilled or deionized water, swirl over 5 min, allow the resin to settle, and then decant the supernatant solution. Pour the resin onto nylon filter cloth on a strainer and wash the resin several times with distilled or deionized water until the pH is 4–7. The Amberlite FPA53 resin as purchased has been converted to the OH^- form before shipping and is ready to use. If alternative resins are used, determine that carbohydrates are not retained by the resin by preparing a test solution consisting of 1 mL 100 mg/mL internal standard and 2.5 mL 10 mg/mL fructo-oligosaccharides diluted to 10 mL. Proceed to **H**(**b**). Recovery of the internal standards and fructo-oligosaccharides should match that of the solution injected directly onto the LC system.

(**p**) *D-Sorbitol (stock internal standards for Sugar-Pak system), 100 mg/mL.*—Dry analytical grade high-purity (>99.5%) D-sorbitol in a freeze-drier at 60°C for 1 day. Weigh 100.00 g D-sorbitol into a beaker, dissolve in water, transfer to a 1 L volumetric flask with water, and dilute to volume. Transfer to a polypropylene bottle and add 0.2 g sodium azide as a preservative. (*Note*: Handle sodium azide with caution only after reviewing MSDS, using appropriate personal protective gear and laboratory hood.) Seal well. Stable at room temperature for >1 year. If the TSK size exclusion chromatographic system is used, replace D-sorbitol with diethylene glycol.

(**q**) *D-Sorbitol (working internal standard), 10 mg/mL.*—Pipet 100 mL solution, **C**(**p**), to a 1 L volumetric flask and dilute to volume with deionized water. Add 0.2 g sodium azide as a preservative. (*Note*: Handle sodium azide with caution only after reviewing MSDS, using appropriate personal protective gear and laboratory hood.) Stable at room temperature for >1 year. If the TSK size exclusion chromatographic system is used, replace D-sorbitol with diethylene glycol.

(**r**) *D-Glucose LC standards (5, 10, 20 mg/mL).*—Accurately weigh 0.5, 1.0, and 2.0 g portions of high-purity (>99.5%) D-glucose (Sigma Chemical Co., St. Louis, MO) and transfer to three separate 100 mL volumetric flasks. To each flask pipet 10 mL internal standard, **C**(**q**). Dilute to volume with 0.02% sodium azide solution, **C**(**t**). Transfer solutions to 100 mL Duran® glass bottles. Stable at room temperature for 1 year.

(**s**) *Deionized water containing $Na_2CaEDTA$ (50 mg/L).*

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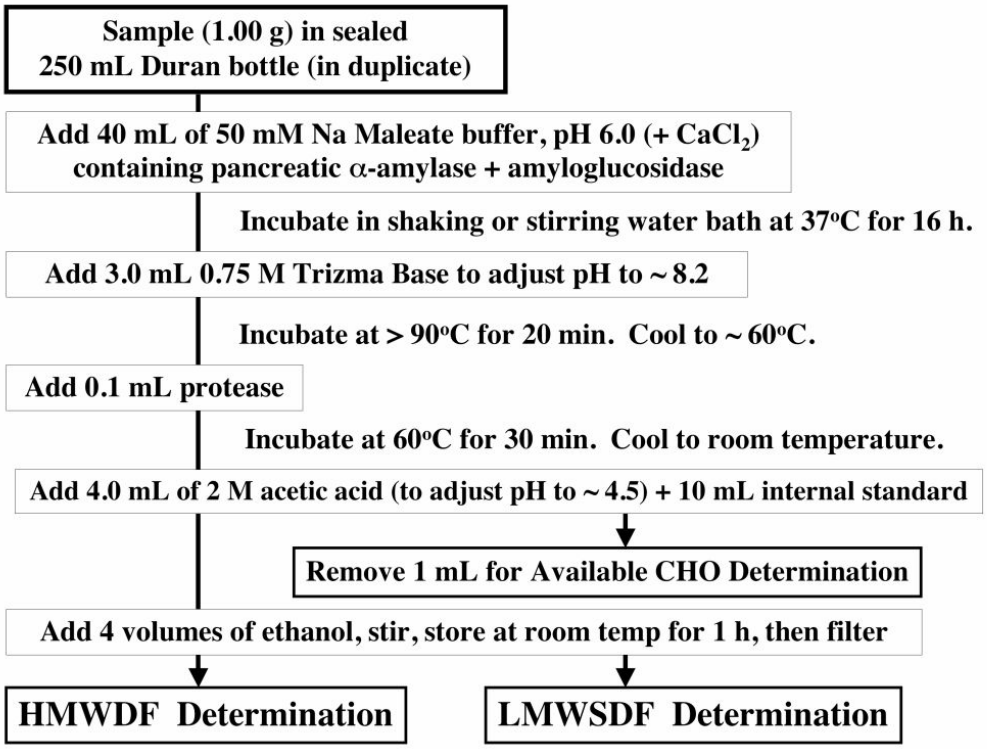

Figure 2009.01B. Schematic representation of the integrated total dietary fiber procedure, also showing where samples can be removed for determination of available carbohydrates (not part of this study).

(**t**) *Sodium azide solution (0.02%, w/v)*.—Add 0.2 g sodium azide to 1 L deionized water and dissolve by stirring. (*Note*: Handle sodium azide with caution only after reviewing MSDS, using appropriate personal protective gear and laboratory hood.) Stable at room temperature for >1 year.

Items (**d**)–(**f**) are supplied in the Integrated Total Dietary Fiber Assay Kit (K-INTDF) available from Megazyme International Ireland Ltd, Bray, Co. Wicklow, Ireland, but preparations of enzymes which meet the criteria as specified in the method may also be used.

D. Preparation of Test Samples

Collect and prepare samples as intended to be eaten, i.e., baking mixes should be prepared and baked, pasta should be cooked, etc. Defat according to AOAC Method **985.29** if >10% fat. For high-moisture samples (>25%) it may be desirable to freeze-dry. Grind ca 50 g in a grinding mill, **B**(**a**), to pass a 0.5 mm sieve. Transfer all material to a wide-mouthed plastic jar, seal, and mix well by shaking and inverting. Store in the presence of a desiccant.

E. Enzyme Purity

To ensure absence of undesirable enzymatic activities and effectiveness of desirable enzymatic activities, run standards listed in Table **991.43B** each time enzyme lot changes or at a maximum 6 month interval.

F. Enzymatic Digestion of Sample

(**a**) *Blanks*.—With each assay, run two blanks along with samples to measure any contribution from reagents to residue.

(**b**) *Samples*.

(*1*) *Duplicate samples*.—1.000 ± 0.005 g samples accurately into Duran glass bottles.

(*2*) *Addition of enzymes*.—Wet the sample with 1.0 mL ethanol (or IMS) and add 40 mL pancreatic α-amylase/AMG mixture, **C**(**e**), to each bottle. Cap the bottles. Transfer the bottles to a Grant OLS 200 shaking incubation bath (or similar) and secure the bottles in place with the springs in the shaker frame.

(*3*) *Incubation with pancreatic α-amylase/AMG*.—Incubate the reaction solutions at 150 revolutions/min (orbital motion) or the relevant rate to ensure suspension in a reciprocal shaker at 37°C for exactly 16 h (e.g., 5.00 p.m. to 9.00 a.m.).

(*4*) *Adjustment of pH to approximately 8.2 (pH 7.9–8.4); inactivation of α-amylase and AMG*.—After 16 h, remove all sample bottles from the shaking water bath and immediately add 3.0 mL 0.75 M Trizma base solution to terminate the reaction. At the same time, if only one shaker bath is available, increase the temperature of the shaking incubation bath to 60°C in readiness for the protease incubation step. Slightly loosen the caps of the sample bottles, place the bottles in a water bath (nonshaking) at 95–100°C, and incubate for 20 min with occasional shaking by hand. Using a

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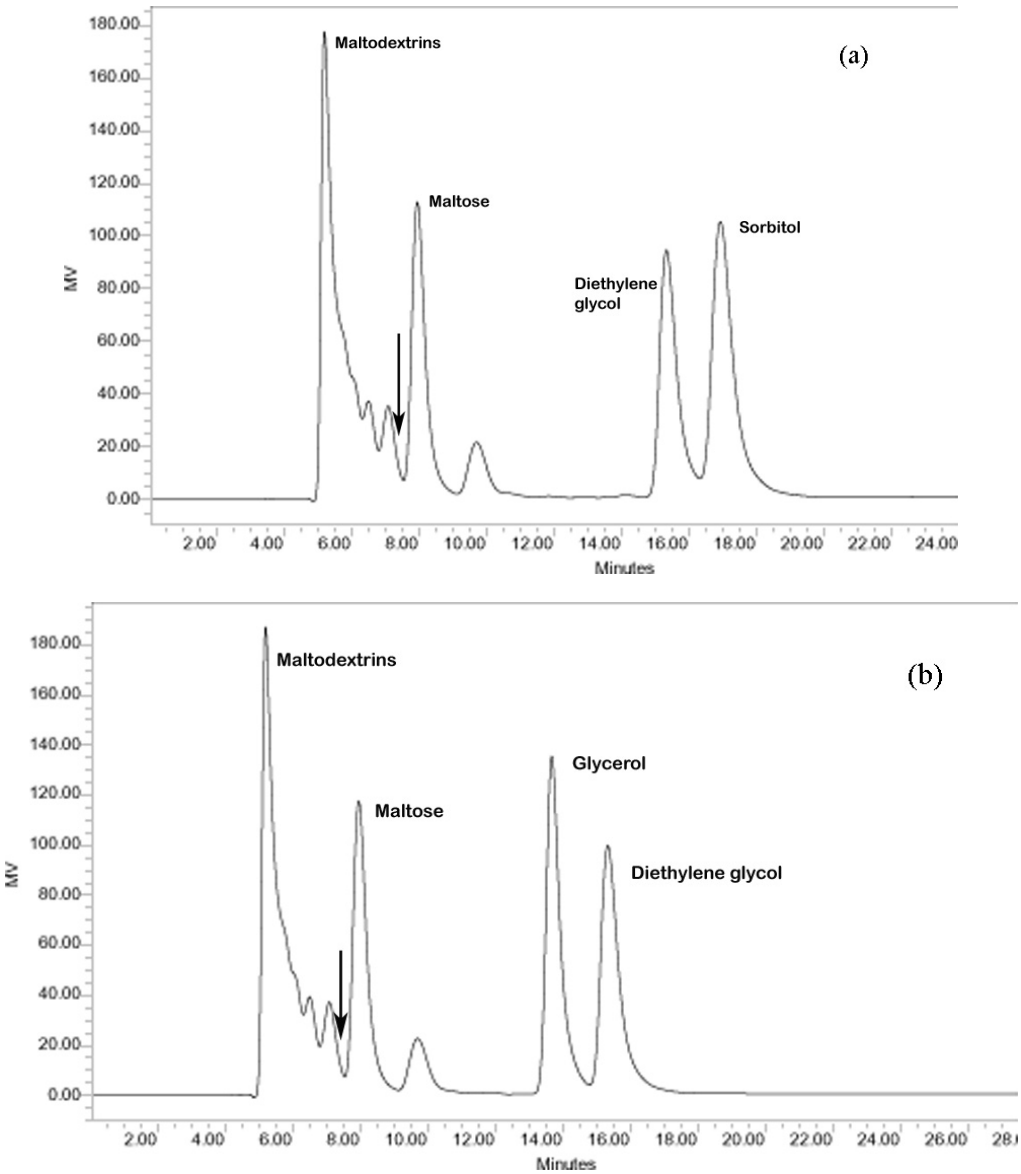

Figure 2009.01C. Chromatography of a mixture of maltodextrins, maltose, diethylene glycol plus either (a) D-sorbitol or (b) glycerol on a Waters Sugar-Pak column. Arrows show demarcation between DP 2 (maltose) and DP 3 (higher maltodextrins).

thermometer, ensure that the final temperature of the bottle contents is >90°C (checking of just one bottle is adequate).

(*5*) *Cool.*—Remove all sample bottles from the hot water bath (use appropriate gloves) and cool to approximately 60°C.

(*6*) *Protease treatment*.—Add 0.1 mL protease solution, **C**(**f**), with a positive displacement dispenser (solution is viscous). Incubate at 60°C for 30 min.

(*7*) *pH adjustment.*—Add 4.0 mL 2 M acetic acid to each bottle, and mix. This gives a final pH of approximately 4.3.

G. Determination of HMWDF

(**a**) *Precipitation of high-molecular-weight soluble dietary fiber (HMWSDF)*.—To each sample, add 10 mL 10 mg/mL internal standard solution, **C**(**q**); then add 227.5 mL (measured at room temperature) of 95% (v/v) EtOH (or IMS), **C**(**a**), preheated to 60°C, and mix thoroughly. Allow the precipitate to form at room temperature for 60 min.

(**b**) *Filtration setup*.—Tare crucible containing Celite to nearest 0.1 mg. Wet and redistribute the bed of Celite in the crucible, using 15 mL 78% (v/v) EtOH (or IMS), **C**(**b**), from wash bottle. Apply suction to crucible to draw Celite onto fritted glass as an even mat.

(**c**) *Filtration*.—Using vacuum, filter precipitated enzyme digest, **G**(**a**), through crucible. Using a wash bottle with 78% (v/v) EtOH or IMS, **C**(**b**), quantitatively transfer all remaining particles to crucible. Retain filtrate and washings [steps (**c**) and (**d**)] for determination of low-molecular-weight soluble dietary fiber (LMWSDF), **H**(**a**).

(**d**) *Wash*.—Using a vacuum, wash residue successively with two 15 mL portions of the following: 78% (v/v) EtOH or IMS; 95% (v/v) EtOH or IMS; acetone.

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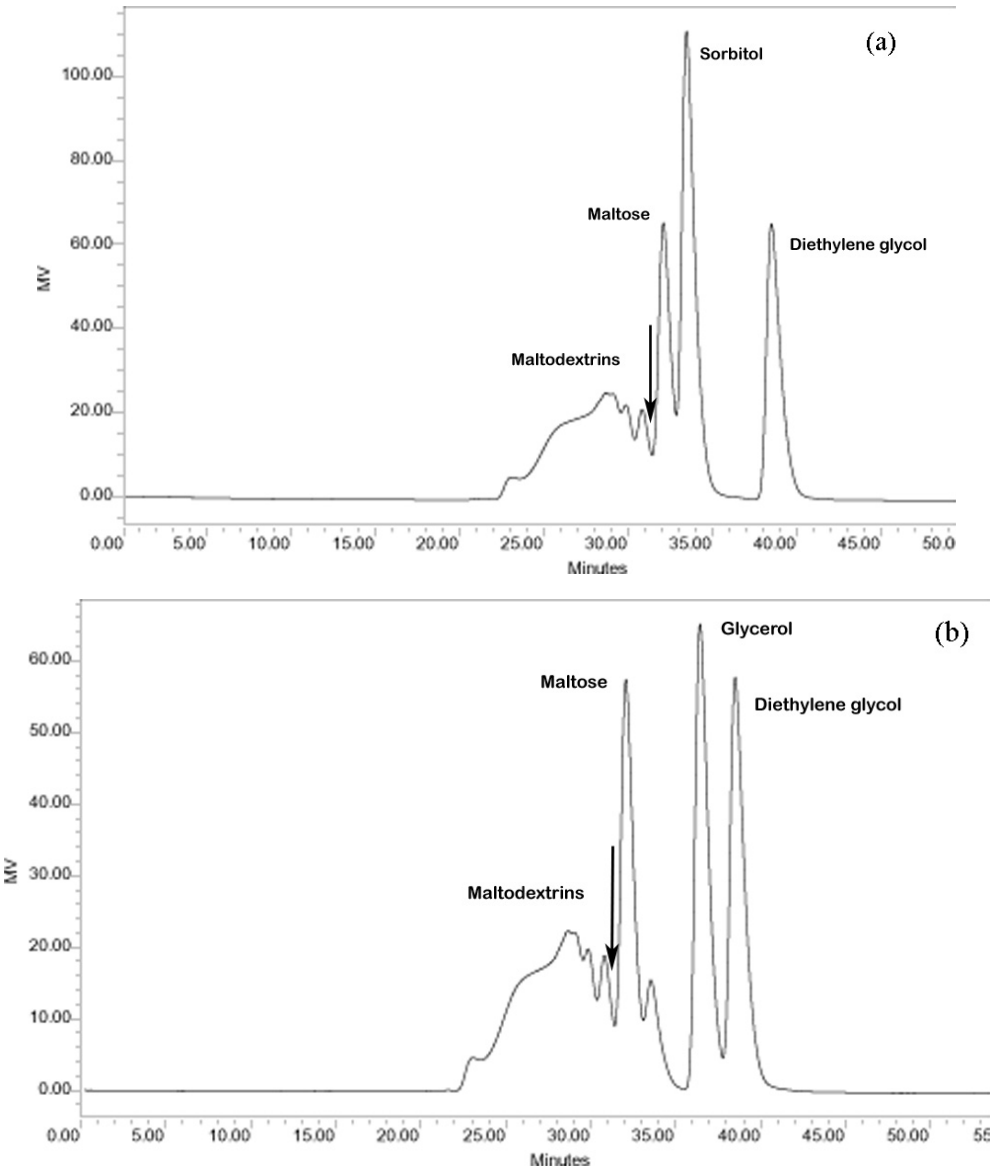

Figure 2009.01D. Chromatography of a mixture of maltodextrins, maltose, diethylene glycol plus either (a) D-sorbitol or (b) glycerol on two TSK-Gel filtration columns (G2500PWXL) in series. Arrows show demarcation between DP 2 (maltose) and DP 3 (higher maltodextrins).

(**e**) *Dry crucibles containing residue overnight in 105°C oven.*

(**f**) *Cool crucible*.—Cool in desiccator for approximately 1 h. Weigh crucible containing dietary fiber residue and Celite to nearest 0.1 mg. To obtain residue mass, subtract tare weight, i.e., weight of dried crucible and Celite.

(**g**) *Protein and ash determination*.—Analyze the residue from one crucible for protein, and the second residue of the duplicate for ash. Perform protein analysis on residue using Kjeldahl or combustion methods. (Exercise caution when using a combustion analyzer for protein in the residue. Celite volatilized from the sample can clog the transfer lines of the unit.) Use 6.25 factor for all cases to calculate mg of protein. For ash analysis, incinerate the second residue for 5 h at 525°C. Cool in desiccator and weigh to nearest 0.1 mg. Subtract crucible and Celite weight to determine ash.

H. Determination of LMWSDF

Note: Proper deionization is an essential part of obtaining quality chromatographic data on LMWSDF. To obtain familiarity regarding the appearance of salt peaks in the LMWSDF chromatograms, dissolve 10 mg sodium chloride into 10 mL 10 mg/mL LC internal standard, **C**(**q**), and proceed to **H**(**c**) at "Transfer…to a 10 mL disposable…". To ensure the resins being used are of adequate deionizing capacity, dissolve 50 mg sodium chloride in 10 mL deionized water. Add 10 mL 10 mg/mL LC internal standard, **C**(**q**), and proceed to (**b**) at "Quantitatively transfer…to a column…". The liquid chromatogram of this solution should show no peaks in the time range corresponding to carbohydrates of DP ≥3.

(**a**) *Filtrate recovery, deionization, and LC analysis*.—Set aside the filtrate from one of the sample duplicates, **G**(**d**), to

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use in case of spills or if duplicate LMWDF data is desired. Transfer approximately one-half of filtrate, **G**(**d**), of the other sample duplicate to a 1 L evaporator flask and concentrate with a rotary evaporator to near dryness at 50°C. Repeat with remaining half of filtrate.

(**b**) *Deionization of sample.*—Dissolve the residue in approximately 20 mL deionized water, and transfer quantitatively to a sealable container if planning to store overnight before deionization. Quantitatively transfer to a column (25 cm × 2.5 cm id), **C**(**o**), containing 25 g each, thoroughly mixed, Amberlite FPA 53 (OH^-), **C**(**o**)(*1*), and Ambersep 200 (H^+), **C**(**o**)(*2*), prepared just before use, collecting the eluate (eluting at a consistent rate of 0.5 to 2 mL/min) into a 500 mL round-bottom rotary evaporator flask. Continue to elute extract through the column with 200 mL deionized water at a consistent rate of 0.5–2.0 mL/min. Evaporate to near dryness at 50°C.

(**c**) *Preparation of samples for LC and LC analyses.*—Quantitatively transfer the concentrate to a 10 mL volumetric flask (using a few milliliters of deionized water to rinse the flask) and dilute to volume (10 mL) with deionized water. Transfer the contents of the 10 mL volumetric flask to a 10 mL disposable syringe, **B**(**bb**), and filter through a 0.45 µm filter, **B**(**y**). Use a 100 µL LC glass syringe, **B**(**cc**), to fill the 50 µL injection loop on the LC, **B**(**t**). Perform this analysis in duplicate.

(**d**) *Determine the response factor for* D*-glucose*.—Because D-glucose provides an LC RI response equivalent to the response factor for the nondigestible oligosaccharides that make up LMWSDF, the LC is calibrated using D-glucose, and the response factor is used for determining the mass of LMWSDF. Use a 100 µL LC syringe, **B**(**cc**), to fill the 50 µL injection loop for each standard internal standard/D-glucose solution, **C**(**r**). Inject in triplicate.

(*1*) *Internal standard method.*—Obtain the values for the peak areas of D-glucose and internal standard from the three chromatograms. The reciprocal of the slope obtained by comparing the ratio of peak area of D-glucose/peak area of internal standard (*y*-axis) to the ratio of the mass of D-glucose/mass of internal standard (*x*-axis) is the response factor. Determine the average response factor (typically 0.91 for D-sorbitol, or 0.73 for diethylene glycol).

$$\text{Response factor (Rf)} = (\text{PA-IS})/(\text{PA-Glu}) \times (\text{Wt-Glu}/\text{Wt-IS})$$

where PA-Glu = peak area D-glucose; PA-IS = peak area internal standard; Wt-Glu = mass of D-glucose in standard; Wt-IS = mass of internal standard in standard.

(*2*) *External standard method*.—Obtain the values for the peak areas of D-glucose from the three chromatograms. Determine the average response factor:

$$\text{Response factor (Rf)} = (\text{Wt-Glu})/(\text{PA-Glu})$$

where PA-Glu = peak area D-glucose; Wt-Glu = mass of D-glucose in standard.

(**e**) *Calibrate the area of chromatogram to be measured for LMWSDF.*—Use a 100 µL LC syringe, **B**(**cc**), to fill the 50 µL injection loop with retention time standard, **C**(**n**). Inject in duplicate. Determine demarcation point between DP 2 and DP 3 oligosaccharides (disaccharides sucrose and maltose vs higher oligosaccharides). *See* Figures **2009.01C**(**a**) and (**b**) and **2009.01D**(**a**) and (**b**).

(**f**) *Determine peak area of LMWSDF (PA LMWSDF) and internal standard (PA internal standard) in chromatograms of sample extracts*.—Inject sample extracts, **H**(**c**), on LC system. Record area of all peaks of DP greater than the DP 2/DP 3 demarcation point as PA LMWSDF. Record the peak area of internal standard as PA-IS.

I. Calculations for HMWDF

Blank (B, mg) determination:

$$B = [(BR1 + BR2)]/2 - P_B - A_B$$

where BR1 and BR2 = residue mass (mg) for duplicate blank determinations, respectively, and P_B and A_B = mass (mg) of protein and ash, respectively, determined on first and second blank residues.

HMWDF:

$$\text{HMWDF (mg/100 g)} = [(R1 + R2)/2 - P - A - B]/(M1 + M2)/2] \times 100$$

where R1 = residue mass 1 from M1 in mg; R2 = residue mass 2 from M2 in mg; M1 = test portion mass 1 in g; M2 = test portion mass 2 in g; A = ash mass from R1; P = protein mass from R2.

J. Calculations for LMWSDF

(*1*) *Internal standard method.*

$$\text{LMWSDF (mg/100 g)} = \text{Rf} \times (\text{Wt-IS, mg}) \times (\text{PA LMWSDF})/(\text{PA-IS}) \times 100/\text{M}$$

where Wt-IS is mg of internal standard contained in 10 mL internal standard solution pipetted into sample filtrate; PA LMWSDF is the peak area of the low-molecular-weight soluble dietary fiber; PA-IS is the peak area of the internal standard; and M is the test portion mass M1 or M2 of the sample whose filtrate was concentrated and analyzed by LC.

(*2*) *External standard method.*

$$\text{LMWSDF (mg/100 g)} = \text{Rf} \times (\text{PA LMWSDF}) \times 100/\text{M}$$

where PA LMWSDF is the peak area of the low-molecular-weight soluble dietary fiber; M is the test portion mass M1 or M2 of the sample whose filtrate was concentrated and analyzed by LC.

K. Calculation of TDF

$$\text{TDF, \%} = (\text{HMWDF} + \text{LMWSDF})/1000$$

Reference: *J. AOAC Int.* **93**, 221 (2010).

Results and Discussion

To simulate food digestion in the small intestine, a combination of gentle shaking combined with enzymatic

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digestion at 37°C is used. Temporary heating to 100°C destroys the amylase and amyloglucosidase activity and promotes some denaturation of protein, providing efficient protein digestion after cooling to 60°C. Incorporating the high-molecular-weight dietary fiber segregation steps and the low-molecular-weight dietary fiber quantitation from previously adopted *Official Methods of Analysis* completes the assay.

The raw data results of the dietary fiber collaborative study are shown in Table 3. Cochran and Grubbs' outliers are noted in the table. Table **2009.01** shows the statistical results obtained after removal of outliers as described above. As stated, samples for this collaborative study were chosen to be challenging, i.e., with emphasis on quantitating products high in RS (legumes, RS ingredient, and whole grain products) and products with typical levels of nondigestible oligosaccharides (all samples). Results show that the within-laboratory variability s_r ranged from 0.41 to 1.43, and the between-laboratory variability s_R ranged from 1.18 to 5.44. Again, as with the precollaborative results, when compared to the dietary fiber statistical results in Table 2, this level and range of variability are similar to those of other dietary fiber methods, influenced by the significant number of technique-dependent manual operations, each of which contributes to the overall variability of the final results.

Collaborators' Comments

One collaborator's comment was that LMWSDF results were slightly higher when calculated by the external standard method than when the internal standard method with sorbitol used as the internal standard. The collaborator believes this is due, in part, to the difficulty of properly integrating the peak area of the sorbitol because it is not always baseline-resolved on both sides. The Study Directors understand this problem and are seeking an alternative internal standard that does not demonstrate these deficiencies; however, they do not believe this had a significant impact on the results of this study.

Another collaborator's comment concerned the use of Duran bottles rather than beakers for the enzymatic digestion steps of the method. The cap on the Duran bottles retains liquid that must be carefully recovered during filtering. It is also more difficult to transfer traces of dietary fiber from the neck of the bottle to the filter than it is with a beaker, as used in AOAC Methods **985.29** and **991.43**. Duran bottles were chosen, in conjunction with the shaker, to ensure that the entire sample being tested properly contacts the enzymes during digestion. Use of the bottle and shaker does not allow rings of sample to form above the solution and keeps the mix homogeneous during the entire digestion.

Recommendations

The Study Directors recommend that this method be adopted by AOAC INTERNATIONAL as Official First Action.

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